

Judge transport, turbulence, and the mean state together

One controlled intervention, three complementary tests of physical behavior.
No completed SurfaceLayerDiffusivity LES result is being claimed here.

01 Did the near-wall transport change as intended?

Lead with resolved, added, and combined momentum/heat flux profiles.
Zoom the lowest four levels; mark every active face. Pair flux mismatch with viscosity, diffusivity, guard activity, and mean vertical transport.

02 Did the boundary layer improve without losing turbulence?

Place mean wind and temperature beside w^2 , w^3 and skewness. Show boundary-layer depth and surface flux through time. Retain the fixed 1 m GABLS1 median on comparable panels; gray bands show archive spread.

03 Is the result robust to filter time and support?

Use the same four styles in every GABLS panel. Compare with a freshly run, same-revision control, then examine 7-8 h versus 8-9 h. Report changes in physical units; avoid a composite score that hides tradeoffs.

-  Matched WENO9 control
-  One face / 100 s
-  One face / 300 s
-  Two faces / 300 s

Independent challenges strengthen the conclusion

GABLS3: moist nocturnal evolution and the sunrise flux sign change.
Neutral ABL: fixed-stress transport, without a surface-drag prediction.
Both add evidence; neither substitutes for the GABLS1 reference comparison.